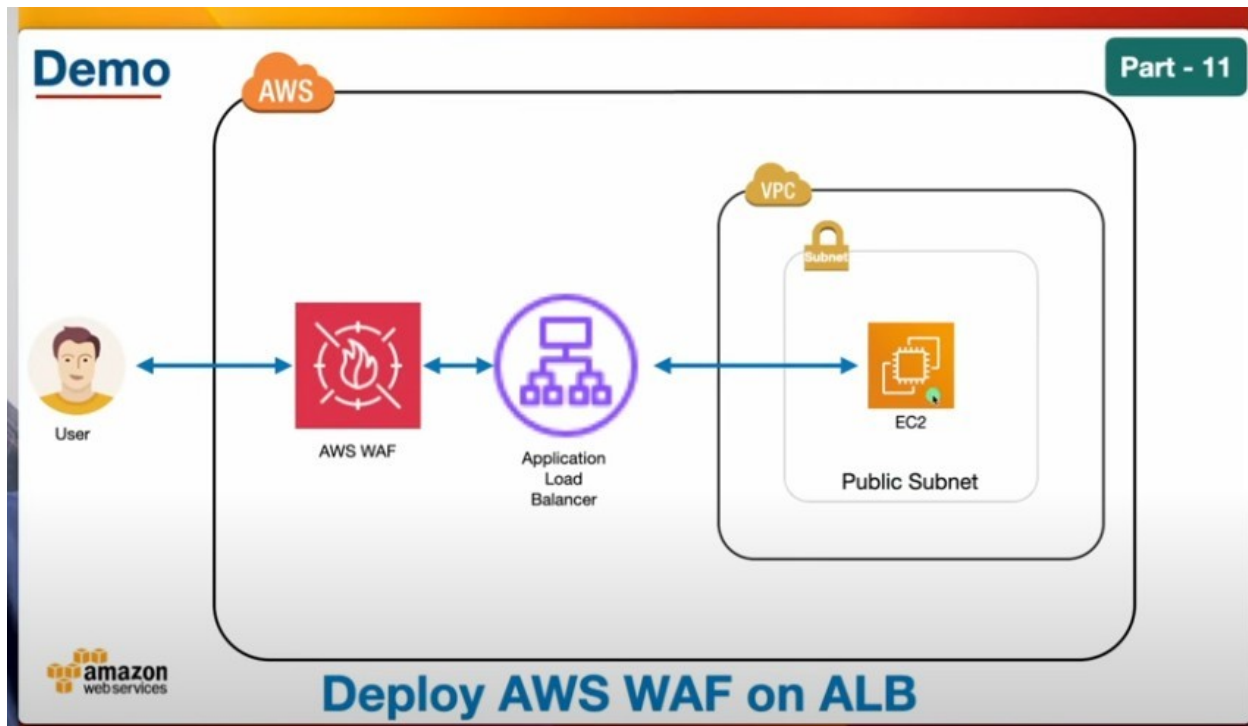


AWS WAF (Web application firewall)/Web ACL

LAB



STEP1: CREATE VPC

VPC > Your VPCs > Create VPC

Create VPC

Info

A VPC is an isolated portion of the AWS Cloud populated by AWS objects, such as Amazon EC2 instances.

VPC settings

Resources to create

Info

Create only the VPC resource or the VPC and other networking resources.

☒ VPC only

☐ VPC and more

Name tag - optional

Creates a tag with a key of 'Name' and a value that you specify.

test-vpc

IPv4 CIDR block

Info

☒ IPv4 CIDR manual input

☐ IPAM-allocated IPv4 CIDR block

IPv4 CIDR

12.0.0.0/16

CIDR block size must be between /16 and /28.

IPv6 CIDR block

Info

☒ No IPv6 CIDR block

☐ IPAM-allocated IPv6 CIDR block

☐ Amazon-provided IPv6 CIDR block

☐ IPv6 CIDR owned by me

Tenancy

Info

Default

Tags

A tag is a label that you assign to an AWS resource. Each tag consists of a key and an optional value. You can use tags to search and filter your resources or track your AWS costs.

Key

Q Name

X

Value - optional

Q test-vpc

X

Add tag

You can add 49 more tags

Remove tag

Cancel

Preview code

Create VPC

VPC > Your VPCs > vpc-026e38639fdcf617

VPC dashboard

EC2 Global View

Filter by VPC

Virtual private cloud

Your VPCs

Subnets

Route tables

Internet gateways

Egress-only internet gateways

DHCP option sets

Elastic IPs

Managed prefix lists

NAT gateways

Peering connections

Route servers

Security

Network ACLs

Security groups

PrivateLink and Lattice

Getting started

Endpoints

Endpoint services

You successfully created vpc-026e38639fdcf617 / test-vpc

vpc-026e38639fdcf617 / test-vpc

Actions

Details

Info

VPC ID

vpc-026e38639fdcf617

DNS resolution

Enabled

Main network ACL

acl-0c924cf6220764b4

IPv6 CIDR

-

State

Available

Tenancy

default

Default VPC

No

Network Address Usage metrics

Disabled

Block Public Access

Off

DHCP option set

dopt-0f2987a43c2f70ca

IPv4 CIDR

12.0.0.0/16

Route 53 Resolver DNS Firewall rule groups

-

DNS hostnames

Disabled

Main route table

rtb-030c3319ff2549e7

IPv6 pool

-

Owner ID

075091538667

Resource map

CIDRs

Flow logs

Tags

Integrations

Resource map

Info

VPC

Your AWS virtual network

test-vpc

Subnets (0)

Subnets within this VPC

Route tables (1)

Route network traffic to resources

rtb-030c3319ff2549e7

Network Connections (0)

Connections to other networks

Show all details

STEP2: CREATE INTERNET GATEWAY

VPC > Internet gateways > Create internet gateway

Create internet gateway [Info](#)

An internet gateway is a virtual router that connects a VPC to the internet. To create a new internet gateway specify the name for the gateway below.

Internet gateway settings

Name tag

Creates a tag with a key of 'Name' and a value that you specify.

IG-TEST

Tags - optional

A tag is a label that you assign to an AWS resource. Each tag consists of a key and an optional value. You can use tags to search and filter your resources or track your AWS costs.

Key

Value - optional

Q Name X

Q IG-TEST X Remove

Add new tag

You can add 49 more tags.

Cancel

Create internet gateway

aws

Search [Alt+S]

United States (Ohio) Account ID: 0750-9153-8667

VPC > Internet gateways > igw-0557bcbc1fb19aa7e

VPC dashboard

EC2 Global View

Filter by VPC

Virtual private cloud

Your VPCs

Subnets

Route tables

Internet gateways

Egress-only internet gateways

DHCP option sets

Elastic IPs

Managed prefix lists

NAT gateways

Peering connections

The following internet gateway was created: igw-0557bcbc1fb19aa7e - IG-TEST. You can now attach to a VPC to enable the VPC to communicate with the internet.

Attach to a VPC

igw-0557bcbc1fb19aa7e / IG-TEST

Actions

Details

Internet gateway ID

igw-0557bcbc1fb19aa7e

State

Detached

VPC ID

-

Owner

075091538667

Tags

Search tags

Manage tags

Key

Value

Name

IG-TEST

STEP3: NOW ATTATCH INTERNET GATEWAY TO VPC

VPC > Internet gateways > igw-0557bcbc1fb19aa7e

VPC dashboard

EC2 Global View

Filter by VPC

Virtual private cloud

Your VPCs

Subnets

Route tables

Internet gateways

Egress-only internet gateways

DHCP option sets

Elastic IPs

Managed prefix lists

NAT gateways

Peering connections

Route servers

Internet gateway igw-0557bcbc1fb19aa7e successfully attached to vpc-026e38639fdcf617

Notifications

igw-0557bcbc1fb19aa7e / IG-TEST

Actions

Details

Internet gateway ID

igw-0557bcbc1fb19aa7e

State

Attached

VPC ID

vpc-026e38639fdcf617 | test-vpc

Owner

075091538667

Tags

Search tags

Manage tags

Key

Value

Name

IG-TEST

STEP4:NOW CREATE SUBNET

Create subnet [Info](#)

VPC

VPC ID

Creates subnets in this VPC.

vpc-026e38639fdcf617 (test-vpc)

Associated VPC CIDRs

IPv4 CIDRs

12.0.0.0/16

Subnet settings

Specify the CIDR blocks and Availability Zone for the subnet.

Subnet 1 of 2

Subnet name

Create a tag with a key of 'Name' and a value that you specify.

test-public-subnet-2a

The name can be up to 256 characters long.

Availability Zone

[Info](#)

Choose the zone in which your subnet will reside, or let Amazon choose one for you.

United States (Ohio) / use2-az1 (us-east-2a)

IPv4 VPC CIDR block

[Info](#)

Choose the VPC's IPv4 CIDR block for the subnet. The subnet's IPv4 CIDR must lie within this block.

12.0.0.0/16

IPv4 subnet CIDR block

12.0.1.0/24

256 IPs

▼ Tags - optional

Key

Q Name

×

Value - optional

Q test-public-subnet-2a

×

Remove

Add new tag

You can add 49 more tags.

Remove

Subnet 2 of 2

Subnet name

Create a tag with a key of 'Name' and a value that you specify.

test-public-subnet-2b

The name can be up to 256 characters long.

Availability Zone

[Info](#)

Choose the zone in which your subnet will reside, or let Amazon choose one for you.

United States (Ohio) / use2-az2 (us-east-2b)

IPv4 VPC CIDR block

[Info](#)

Choose the VPC's IPv4 CIDR block for the subnet. The subnet's IPv4 CIDR must lie within this block.

12.0.0.0/16

IPv4 subnet CIDR block

12.0.2.0/24

256 IPs

▼ Tags - optional

Key

Q Name

×

Value - optional

Q test-public-subnet-2b

×

Remove

Add new tag

You can add 49 more tags.

Remove

Add new subnet

Add new tag

You can add 49 more tags.

Remove

Add new subnet

Cancel

Create subnet

STEP5: NOW SUBNETS HAS BEEN CRETAEED

STEP6: We need to save association and add route tables

VPC > Route tables > Create route table

Create route table

Info

A route table specifies how packets are forwarded between the subnets within your VPC, the internet, and your VPN connection.

Route table settings

Name - optional

Create a tag with a key of 'Name' and a value that you specify.

RT-test-public

VPC

The VPC to use for this route table.

vpc-026e38639dfc617 (test-vpc)

Tags

A tag is a label that you assign to an AWS resource. Each tag consists of a key and an optional value. You can use tags to search and filter your resources or track your AWS costs.

Key

Q Name

Value - optional

Q RT-test-public

Remove

Add new tag

You can add 49 more tags.

Cancel

Create route table

VPC > Route tables > rtb-0d4012e6604c1898f > Edit subnet associations

Edit subnet associations

Change which subnets are associated with this route table.

Available subnets (2/2)

Q Filter subnet associations

☒	Name	Subnet ID	IPv4 CIDR	IPv6 CIDR	Route table ID
☒	test-public-subnet-2a	subnet-0fd87b69e2bffa10	12.0.1.0/24	-	Main [rtb-030c3]
☒	test-public-subnet-2b	subnet-0d8c1056a0c31cf9d	12.0.2.0/24	-	Main [rtb-030c3]

Selected subnets

subnet-0fd87b69e2bffa10 / test-public-subnet-2a

subnet-0d8c1056a0c31cf9d / test-public-subnet-2b

Cancel

Save associations

STEP7: NOW we need to add route so that the subnet can access the internet

VPC > Route tables > rtb-0d4012e6604c1898f > Edit routes

Edit routes

Destination	Target	Status	Propagated	Route Origin
12.0.0.0/16	local	Active	No	CreateRouteTable
Q 0.0.0.0/0	Internet Gateway	-	No	CreateRoute
	igw-0557bcb1fb19aa7e			Remove

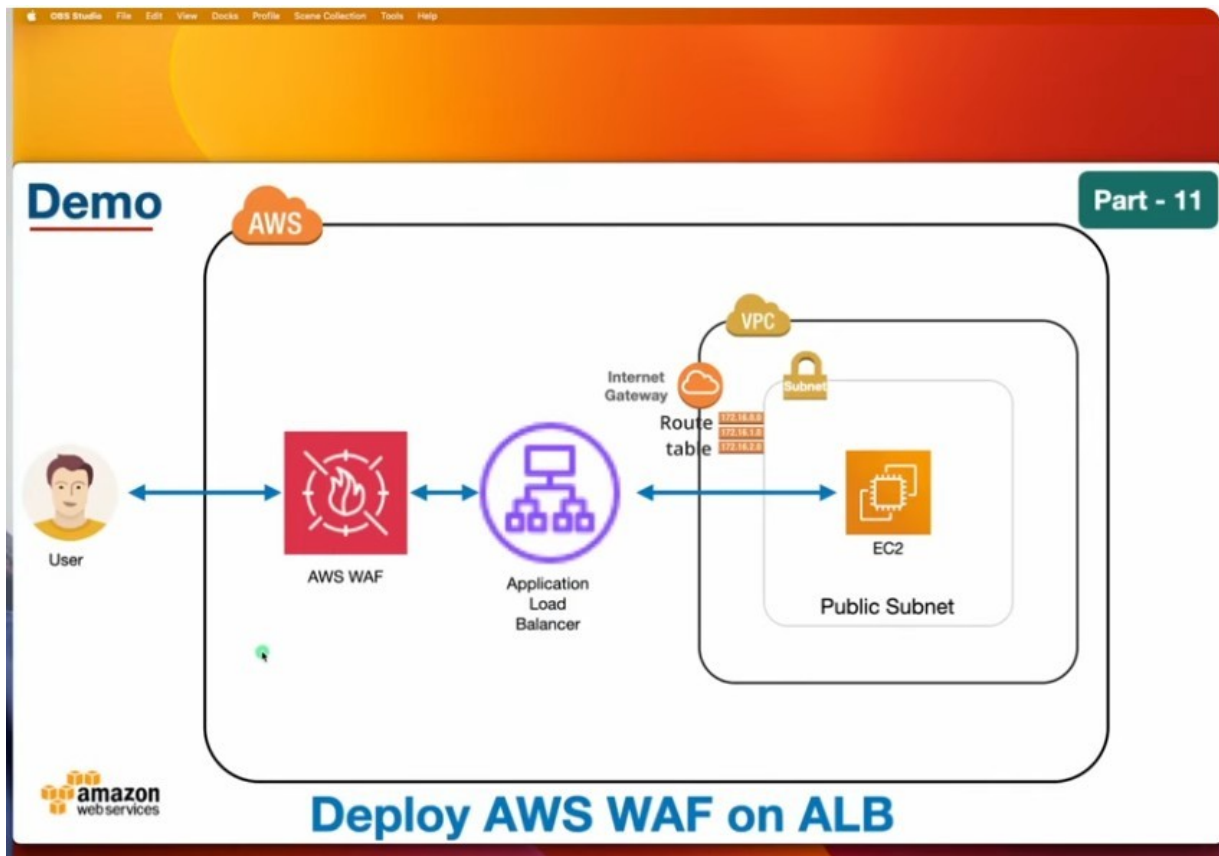
Add route

Cancel

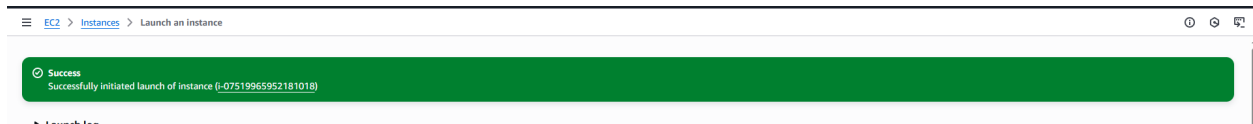
Preview

Save changes

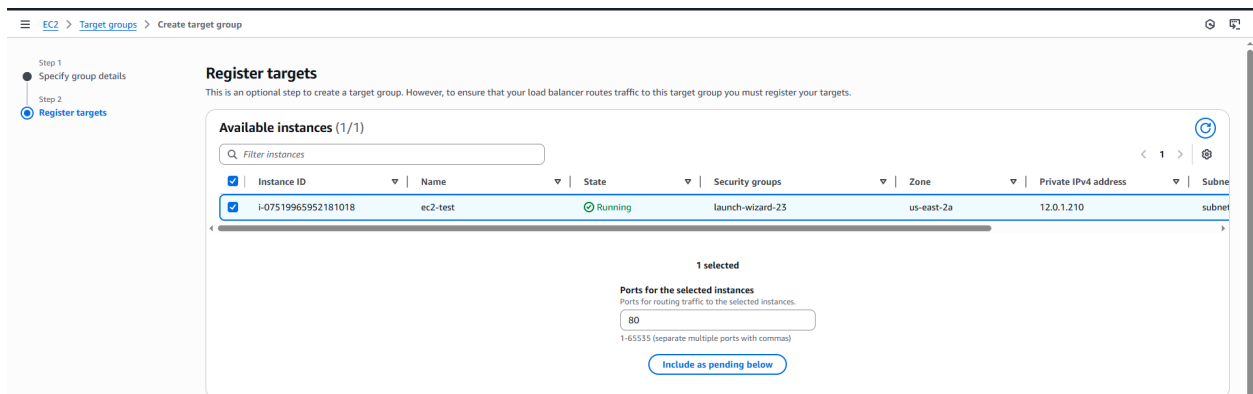
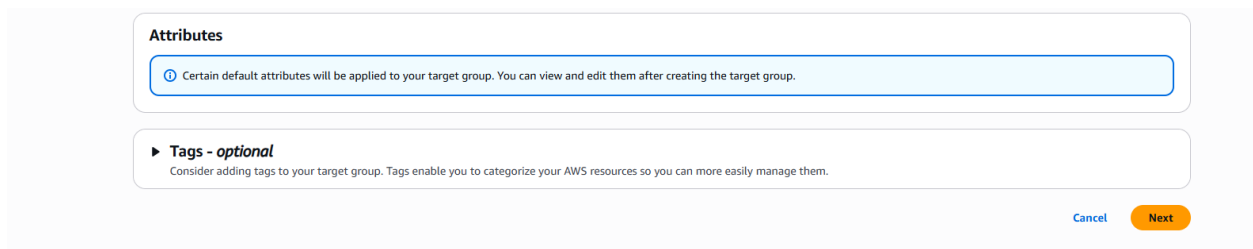
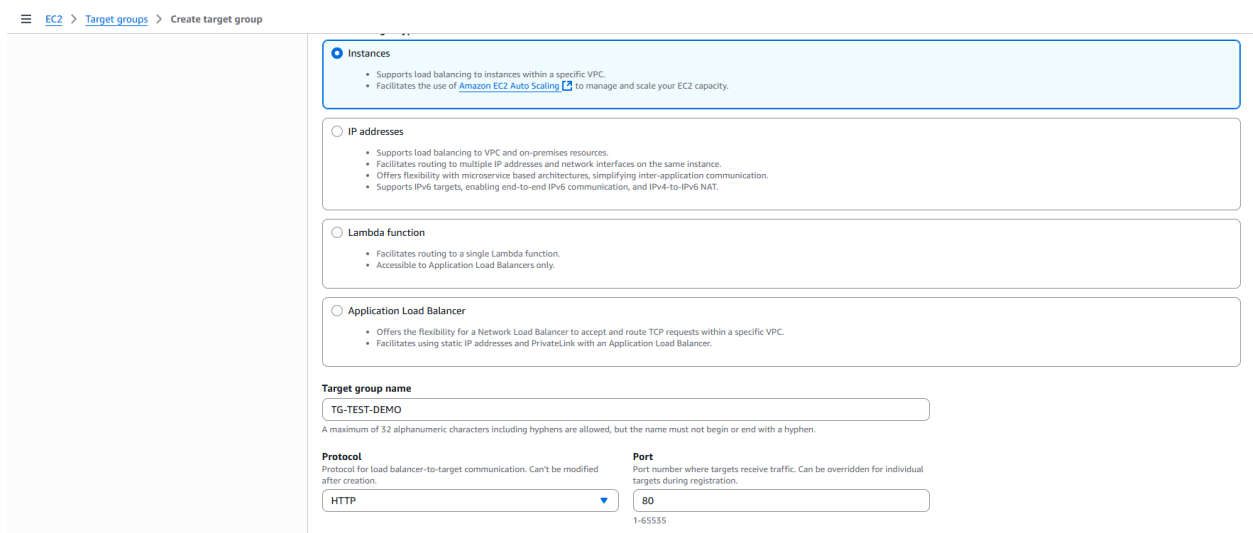
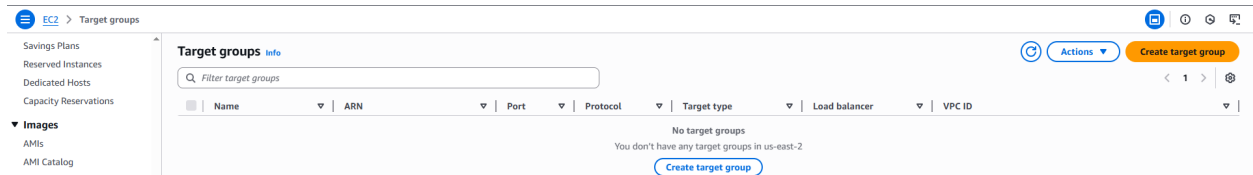
STEP8: STEPS DONE SO FAR



STEP9:NOW CREATE EC2 INSTANCE



STEP10:NOW WE NEED TO CREATE TARGET GROUP



☐ Instance ID

Name

State

Security groups

Zone

Private IPv4 address

Subnet

☐	i-07519965952181018	ec2-test	Running	launch-wizard-23	us-east-2a	12.0.1.210	subnet-
--------------------------	---------------------	----------	---------	------------------	------------	------------	---------

0 selected

Ports for the selected instances

Ports for routing traffic to the selected instances.

1-65535 (separate multiple ports with comma)

1 selection is now pending below. Include more or register targets when ready.

Review targets

Targets (1)

Show only pending

Instance ID	Name	Port	State	Security groups	Zone	Private IPv4 address	Subnet ID	Launch time
i-07519965952181018	ec2-test	80	Running	launch-wizard-23	us-east-2a	12.0.1.210	subnet-0fd87b69e2bffa10	September 9, 2025, 22:53 (UTC-04:00)

1 pending

EC2

Target groups

TG-TEST-DEMO

Savings Plans

Reserved Instances

Dedicated Hosts

Capacity Reservations

Images

AMIs

AMI Catalog

Elastic Block Store

Volumes

Snapshots

Lifecycle Manager

Network & Security

Security Groups

Elastic IPs

Placement Groups

Key Pairs

Network Interfaces

Load Balancing

Load Balancers

Target Groups

Trust Stores

Auto Scaling

Auto Scaling Groups

Settings

Successfully created the target group: TG-TEST-DEMO. Anomaly detection is automatically applied to all registered targets. Results can be viewed in the Targets tab.

TG-TEST-DEMO

Details

Target type

Instance

Protocol : Port

HTTP: 80

Protocol version

HTTP1

VPC

vpc-026e38639dfcf617

IP address type

IPv4

Load balancer

None associated

1

0

0

1

0

0

Total targets

Healthy

Unhealthy

Unused

Initial

Draining

0 Anomalous

Distribution of targets by Availability Zone (AZ)

Select values in this table to see corresponding filters applied to the Registered targets table below.

Targets

Monitoring

Health checks

Attributes

Tags

Registered targets (1)

Anomaly mitigation: Not applicable

Deregister

Register targets

Target groups route requests to individual registered targets using the protocol and port number specified. Health checks are performed on all registered targets according to the target group's health check settings. Anomaly detection is automatically applied to HTTP/HTTPS target groups with at least 3 healthy targets.

Instance ID	Name	Port	Zone	Health status	Health status details	Administrative override	Override details	La
-------------	------	------	------	---------------	-----------------------	-------------------------	------------------	----

STEP11: NOW TARGET GROUP IS CREATED

STEP12: NOW CREATE APPLICATION LOAD BALANCER

For this we need to create security group

STEP12: NOW CREATE APPLICATION LOAD BALANCER

For this we need to create security group

EC2 > Security Groups > Create security group

Inbound rules

Type	Protocol	Port range	Source	Description - optional
HTTP	TCP	80	Anywhere... 0.0.0.0/0	
SSH	TCP	22	Anywhere... 0.0.0.0/0	

Add rule

Rules with source of 0.0.0.0/0 or ::/0 allow all IP addresses to access your instance. We recommend setting security group rules to allow access from known IP addresses only.

Outbound rules

Type	Protocol	Port range	Destination	Description - optional
All traffic	All	All	Custom 0.0.0.0/0	

Add rule

STEP13: ADD THE SECURITY GROUP AND CLICK ON NEXT

EC2 > Load balancers > Create Application Load Balancer

The load balancer will exist and scale within the selected VPC. The selected VPC is also where the load balancer targets must be hosted unless routing to Lambda or on-premises targets, or if using VPC peering. To confirm the VPC for your targets, view [target groups](#).

vpc-026e38639dfcf617 (test-vpc)
12.0.0.0/16

Create VPC

IP pools

You can optionally choose to configure an IPAM pool as the preferred source for your load balancer IP addresses. Create or view Pools in the [Amazon VPC IP Address Manager console](#).

☐ Use IPAM pool for public IPv4 addresses

The IPAM pool you choose will be the preferred source of public IPv4 addresses. If the pool is depleted IPv4 addresses will be assigned by AWS.

Availability Zones and subnets

Select at least two Availability Zones and a subnet for each zone. A load balancer node will be placed in each selected zone and will automatically scale in response to traffic. The load balancer routes traffic to targets in the selected Availability Zones only.

☒ us-east-2a (use2-az1)

Subnet
Only CIDR blocks corresponding to the load balancer IP address type are used. At least 8 available IP addresses are required for your load balancer to scale efficiently.

subnet-0fd87b69e2bffa10
IPv4 subnet CIDR: 12.0.1.0/24

test-public-subnet-2a

☒ us-east-2b (use2-az2)

Subnet
Only CIDR blocks corresponding to the load balancer IP address type are used. At least 8 available IP addresses are required for your load balancer to scale efficiently.

subnet-0dbc1056a0c31cf9d
IPv4 subnet CIDR: 12.0.2.0/24

test-public-subnet-2b

Security groups

A security group is a set of firewall rules that control the traffic to your load balancer. Select an existing security group, or you can [create a new security group](#).

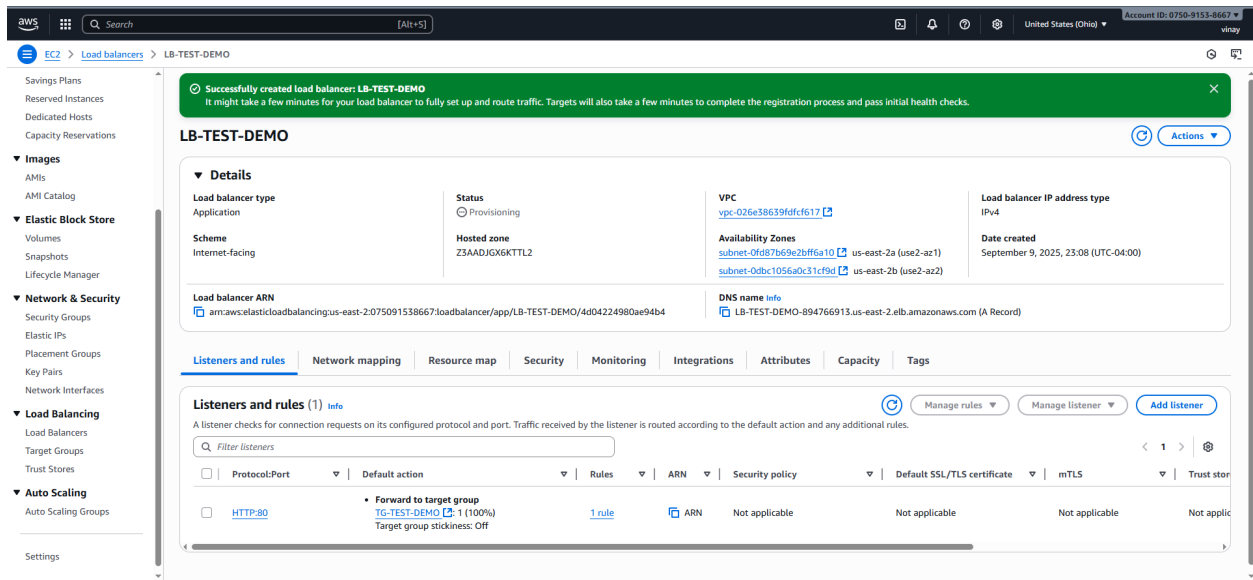
Security groups

Select up to 5 security groups

default sg-081147f0b0e4e24cd VPC: vpc-026e38639dfcf617

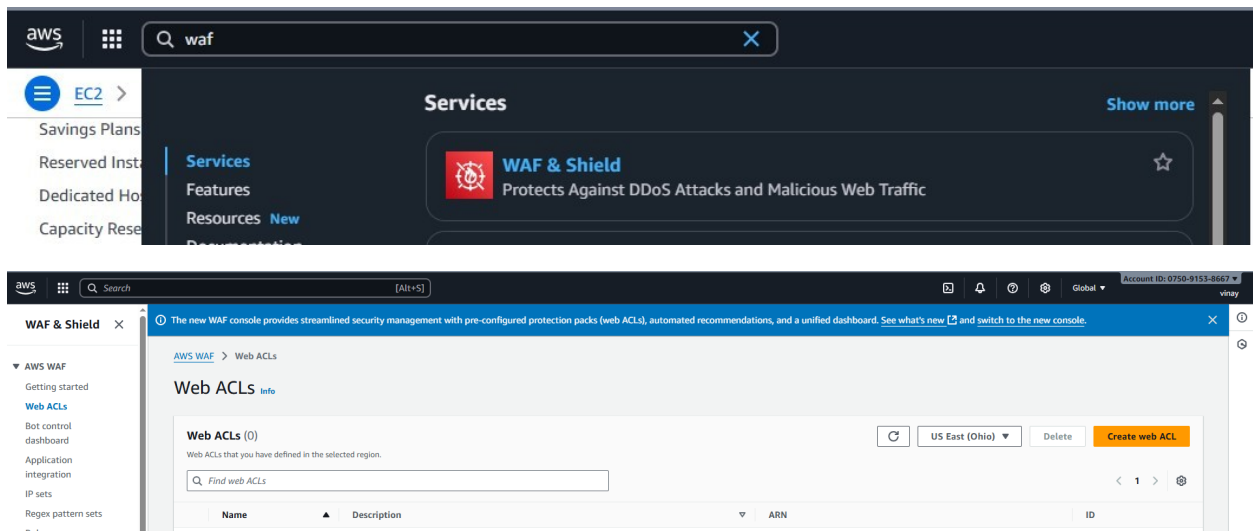
MY SG-PORT-80 sg-0a706149d73ed44a4 VPC: vpc-026e38639dfcf617

STEP14: LOAD BALANCER has been created



STEP15:NOW we need to create WEB APPLICATION FIREWALL.

Go to AWS CONSOLE and SEARCH FOR WAF



STEP16: NOW CLICK on CREATE WEB ACL

associate it to AWS resources

Step 2
Add rules and rule groups

Step 3
Set rule priority

Step 4
Configure metrics

Step 5
Review and create web ACL

Web ACL details

Name

The name must have 1-128 characters. Valid characters: A-Z, a-z, 0-9, - (hyphen), and _ (underscore).

Description - optional

The description can have 1-256 characters.

CloudWatch metric name

The name must have 1-128 characters. Valid characters: A-Z, a-z, 0-9, - (hyphen), and _ (underscore).

Resource type
Choose the type of resource to associate with this web ACL.

☐ Amazon CloudFront distributions

☒ Regional resources (Application Load Balancers, Amazon API Gateway REST APIs, Amazon App Runner services, AWS AppSync GraphQL APIs, Amazon Cognito user pools and AWS Verified Access Instances)

CloudWatch metric name

The name must have 1-128 characters. Valid characters: A-Z, a-z, 0-9, - (hyphen), and _ (underscore).

Resource type
Choose the type of resource to associate with this web ACL.

☐ Amazon CloudFront distributions

☒ Regional resources (Application Load Balancers, Amazon API Gateway REST APIs, Amazon App Runner services, AWS AppSync GraphQL APIs, Amazon Cognito user pools and AWS Verified Access Instances)

Region
Choose the AWS region to create this web ACL in.

STEP17: NOW WE NEED TO ATTATCH THE LOAD BALANCER

Associated AWS resources - optional (0)

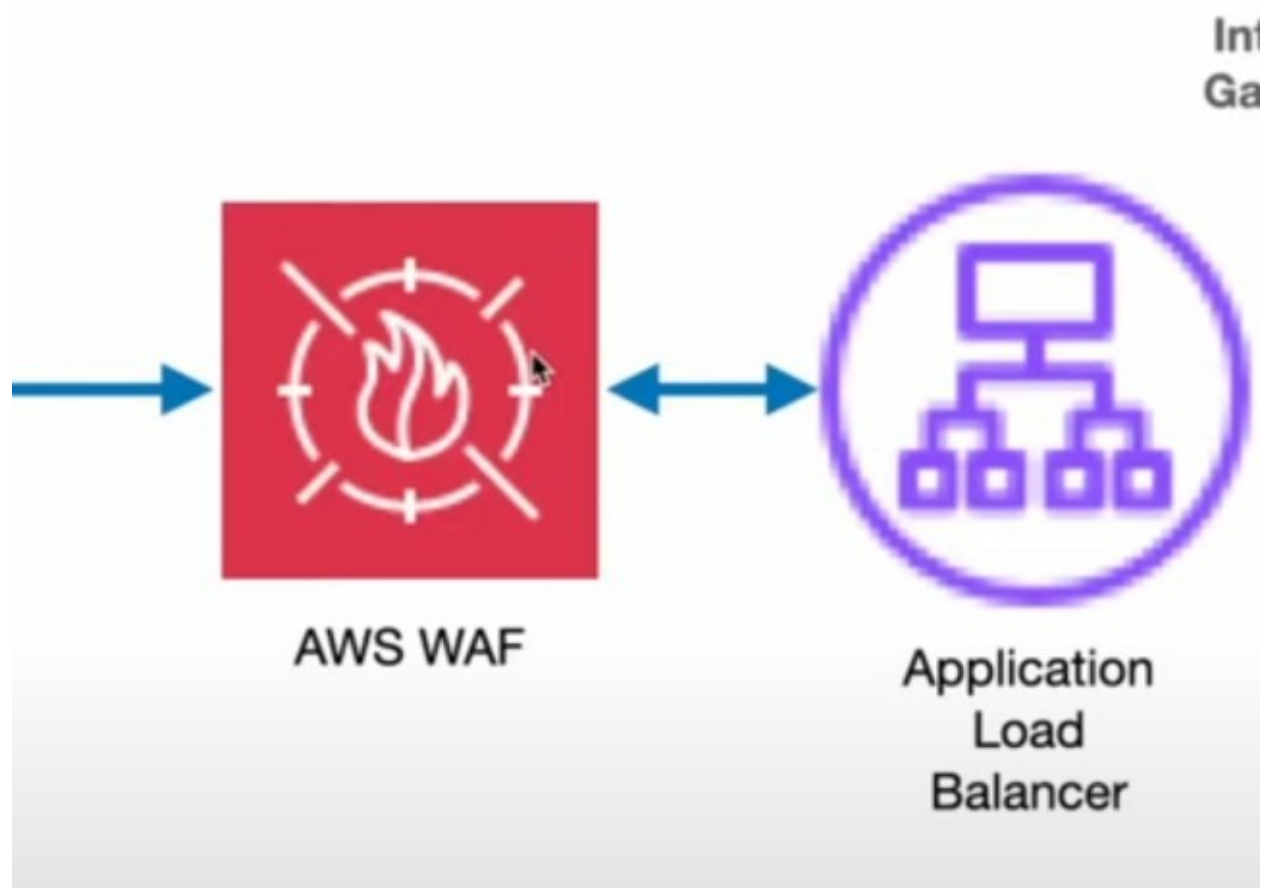
Remove

Add AWS resources

< 1 > ⚙

	Name	Resource type	Region
No items No items to display			

CancelNext



STEP18: NOW add the LOAD BALANCER

Choose the type of resource to associate with this web ACL.

Add AWS resources

Resource type
Select the resource type and then select the resource you want to associate with this web ACL.

☒ Application Load Balancer

☐ Amazon API Gateway REST API

☐ Amazon App Runner service

☐ AWS AppSync GraphQL API

☐ Amazon Cognito user pool

☐ AWS Verified Access

Select the resources you want to associate with the web ACL.

< 1 > ⚙️

☐	Name
☐	lb-aws-waf-demo

Cancel Add

STEP19: NOW we need to add rule groups. This is used to allow or deny access to the EC2 instance

Create web ACL - Step 2 Add

https://us-east-1.console.aws.amazon.com/wafv2/homev2/web-acls/new?region=eu-central-1

AWS Services Search [Option+S] Global rahul.wagh

AWS WAF > Web ACLs > Create web ACL

Step 1
[Describe web ACL and associate it to AWS resources](#)

Step 2
Add rules and rule groups

Step 3
Set rule priority

Step 4
Configure metrics

Step 5
Review and create web ACL

Add rules and rule groups [Info](#)

A rule defines attack patterns to look for in web requests and the action to take when a request matches the patterns. Rule groups are reusable collections of rules. You can use managed rule groups offered by AWS and AWS Marketplace sellers. You can also write your own rules and use your own rule groups.

Rules (0) Edit Delete Add rules

If a request matches a rule, take the corresponding action. The rules are prioritized in order they appear.

	Name	Capacity	Action
	No rules. You don't have any rules added.		

Web ACL capacity units (WCUs) used by your web ACL

The WCUs used by the web ACL will be less than or equal to the sum of the capacities for all of the rules in the web ACL.

The total WCUs for a web ACL can't exceed 5000. Using over 1500 WCUs affects your costs. [AWS WAF Pricing](#)

0/5000 WCUs

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STEp19: To add rules, we need to go to IP SETS

What is an IP Set in AWS?

Simple Explanation

Think of an IP Set in AWS as a "VIP list" or "blacklist" for your website's door guard. AWS WAF (Web Application Firewall) is like the guard that protects your web apps from bad traffic. An IP Set is just a reusable list of IP addresses (like home addresses on the internet) that tells the guard: "Let these in" or "Block those out." It's super straightforward – no fancy tech needed; you group IPs together and plug them into your security rules to control who accesses your site.

In short: IP Sets help you manage groups of IP addresses easily, so you can allow trusted visitors or block suspicious ones without rewriting rules every time.

Detailed Explanation

IP Sets are a feature in AWS WAF, which is a service that helps protect your web applications (like APIs or websites hosted on CloudFront, API Gateway, or Application Load Balancers) from common attacks like SQL injection or DDoS. Specifically, an IP Set is an AWS resource that stores a collection of IP addresses or ranges of IP addresses in CIDR notation (e.g., 192.0.2.0/24 for a block of IPs). These can be IPv4 or IPv6 addresses.

How They Work:

- ▢ You create an IP Set and add IPs to it (single IPs like 203.0.113.1 or ranges like 10.0.0.0/16).
- ▢ Then, in your WAF rules (called "rule statements"), you reference this IP Set. For example:
 - **IP Set Match Statement:** WAF checks if the incoming request's source IP matches anything in your IP Set.
 - If it matches, you can set actions like ALLOW (let it through), BLOCK (stop it), or COUNT (just log it for monitoring).
- ▢ IP Sets are reusable across multiple rules or Web ACLs (Access Control Lists), making it efficient for large-scale security.
- ▢ They support up to thousands of addresses per set (check AWS quotas for exact limits, like 10,000 for standard sets).

- ▢ You can manage them via the AWS Console, CLI, API, or even automate updates with Lambda functions for dynamic blacklists.

Key Features:

- ▢ **Scope:** Choose "Regional" for services like ALB or API Gateway, or "Global" for CloudFront (edge locations).
- ▢ **CIDR Support:** AWS WAF handles /1 to /32 for IPv4 and /1 to /128 for IPv6.
- ▢ **Integration:** Works with other WAF features like rate limiting or managed rules from AWS.
- ▢ **Automation:** You can auto-update IP Sets with AWS IP ranges or third-party feeds (e.g., for blocking known bad actors).

Use Cases:

- ▢ **Whitelisting:** Allow only your office IPs to access an admin panel.
- ▢ **Blacklisting:** Block IPs from spammers, bots, or countries you don't serve (though for geo-blocking, WAF has a separate Geo Match feature).
- ▢ **Incident Response:** Quickly add attacker IPs to a set during a security event.
- ▢ **Compliance:** Restrict access to sensitive apps by IP.

Limitations:

- ▢ Can't use wildcards or hostnames – only IPs/CIDRs.
- ▢ Quotas apply (e.g., max 100 IP Sets per account per region).
- ▢ Doesn't inspect encrypted traffic deeply; it's for IP-level matching.
- ▢ Costs: Charged based on the number of rules and requests processed (pay-as-you-go).

This keeps things modular: Instead of hardcoding IPs into every rule, you update one IP Set, and it applies everywhere.

Mnemonic and Tip for Easy Remembering

Mnemonic: "IP SETS = Internet **P**rotection **S**uper **E**asy **T**ool for **S**ecurity."
Picture a "set" like a Lego set – you build a block of IPs, snap it into your WAF wall, and it fortifies your app instantly.

Tip: To remember, think of it like setting a dinner table: You "set" the IPs (plates) in a group, and WAF decides who sits (allows) or gets kicked out (blocks). Always start small – create one set for testing with your own IP, so you don't lock yourself out!

Step-by-Step Detailed Example: Creating and Using an IP Set to Block Malicious IPs

Let's walk through a real-world example: You're running a web app behind an ALB and want to block traffic from a suspicious IP range (e.g., 203.0.113.0/24). We'll create an IP Set for blocking, add it to a Web ACL rule, and associate with the ALB. I'll assume you're using the AWS Management Console for simplicity (you can also use CLI, SDK, or CloudFormation).

Step 1: Create an IP Set

- ▮ Go to the AWS Management Console > Search for "WAF" > Select "AWS WAF" (make sure you're in the right region, e.g., us-east-1).
- ▮ In the left navigation pane, under "AWS WAF", click "IP sets".
- ▮ Click "Create IP set".
- ▮ **Settings:**
 - Region: Choose "Regional" (for ALB association; use "Global" only for CloudFront).
 - Name: Enter something memorable like "BlockBadIPs".
 - Description: "IP ranges to block for security".
 - IP version: Select "IPv4" (or IPv6 if needed).
 - IP addresses: Add one per line in CIDR format, e.g., "203.0.113.0/24" (this blocks 256 IPs in that range).
- ▮ Click "Create IP set".
- ▮ Tip: Start small—test with your own IP to see blocking in action (but don't lock yourself out!).

Step 2: Create a Web ACL

- ▮ Still in AWS WAF console, in the left pane, click "Web ACLs".
- ▮ Click "Create web ACL".
- ▮ **Settings:**

- Name: "ProtectMyALB".
- Description: "ACL with IP block rule".
- CloudWatch metric name: "ProtectMyALB" (for monitoring).
- Resource type: "Regional resources" (for ALB).
- Associate AWS resources: Skip for now (we'll do this later).
- ▢ Click "Next".
- ▢ On the "Add rules and rule groups" page, click "Add my own rules and rule groups".
- ▢ **Add a Rule:**
 - Rule type: "Regular rule".
 - Name: "BlockBadIPsRule".
 - Priority: 1 (lower number = higher priority; rules are evaluated in order).
 - If a request: "Matches the statement".
 - Statement: Select "IP set" > Choose your IP Set ("BlockBadIPs") > IP address to use: "Source IP address" (or "HTTP header" if using proxies).
 - Action: "Block" (this blocks matching IPs; choose "Allow" for whitelisting).
- ▢ Click "Add rule" > Then "Next" through defaults (no need for managed rules unless you want extras).
- ▢ Review and click "Create web ACL".

Step 3: Associate the Web ACL with Your ALB

- ▢ In the AWS WAF console, go to your new Web ACL ("ProtectMyALB").
- ▢ Click "Associated AWS resources" tab > "Add AWS resources".
- ▢ Select "Application Load Balancer".
- ▢ Choose your ALB from the list (ensure it's in the same region).
- ▢ Click "Add".
- ▢ Alternatively, from the Elastic Load Balancing console:

- o Go to EC2 > Load Balancers > Select your ALB.
- o In the "Integrated services" tab, under "AWS WAF", click "Edit" > Choose your Web ACL > Save.

Step 4: Test and Monitor

- ▮ Send a test request to your ALB from a blocked IP (use a tool like curl or a VPN to simulate).
- ▮ Expected: Blocked requests get a 403 Forbidden response.
- ▮ Monitor: In WAF console, check "Sampled requests" or CloudWatch metrics for blocked/allowed counts.
- ▮ Update: To add/remove IPs, edit the IP Set—it auto-updates the Web ACL.
- ▮ Cleanup Tip: If testing, disassociate the Web ACL first to avoid issues.

Common Pitfalls and Tips

- ▮ **Region Match:** IP Sets and Web ACLs must be in the same region as your ALB.
- ▮ **Order Matters:** Rules in Web ACL are processed top-to-bottom—put critical ones (like IP blocks) high priority.
- ▮ **Fallback Action:** Set the default Web ACL action to "Allow" so non-matching traffic passes through.
- ▮ **Costs:** Minimal—pay per 1,000 requests inspected (~\$0.0006) and per rule (~\$1/month).
- ▮ **Advanced:** For automation, use AWS CLI: e.g., `aws wafv2 create-ip-set` to script IP updates.

This setup is straightforward and scalable. Remember the "IP Shopping List" tip, and you'll never forget! If you need more (e.g., regex or geo-blocking), WAF has those too.

STEP20: NOW create IP SET DETAILS

WAF & Shield ×

AWS WAF

- Getting started
- Web ACLs**
- Bot control dashboard
- Application integration **New**
- IP sets**
- Regex pattern sets
- Rule groups
- AWS Marketplace managed rules

Switch to AWS WAF Classic

[AWS WAF](#) > [Web ACLs](#) > Create web ACL

Step 1
[Describe web ACL and associate it to AWS resources](#)

Step 2
Add rules and rule groups

Step 3
Set rule priority

Step 4
Configure metrics

Step 5
Review and create web ACL

Add rules and rule groups [Info](#)

A rule defines attack patterns to look for in web requests and the action to take when a request matches the patterns. Rule groups are reusable collections of rules. You can use managed rule groups offered by AWS and AWS Marketplace sellers. You can also write your own rules and use your own rule groups.

Rules (0) [Edit](#) [Delete](#) [Add rules ▼](#)

If a request matches a rule, take the corresponding action. The rules are prioritized in order they appear.

	Name	Capacity	Action
No rules. You don't have any rules added.			

Web ACL capacity units (WCUs) used by your web ACL

IP set details

IP set name

my-laptop-ip

The name must have 1-128 characters. Valid characters: A-Z, a-z, 0-9, - (hyphen), and _ (underscore).

Description - optional

The description can have 1-256 characters.

Region

Choose the AWS region to create this IP set in.

Europe (Frankfurt) ▼

IP version

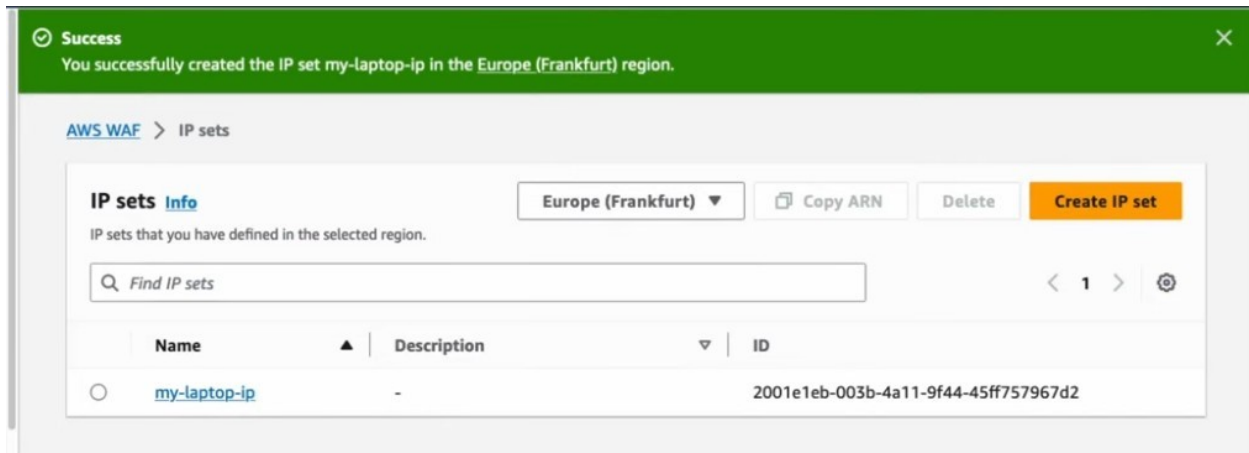
☒ IPv4

☐ IPv6

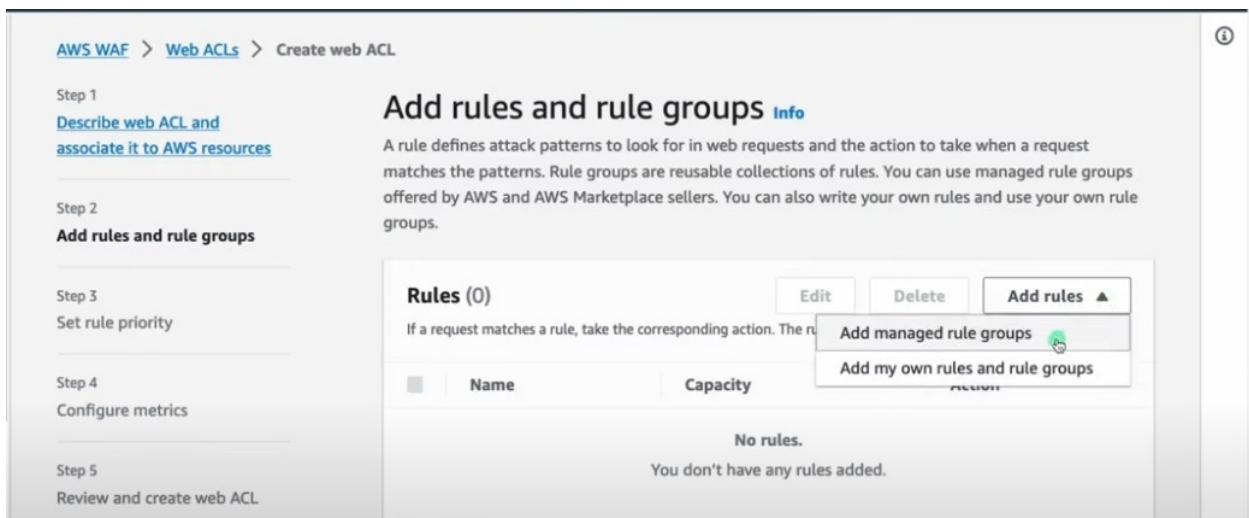
IP addresses

82.199.170.75/32

STEP21: NOW IP SET has been created



STEP22: NOW go to STEP 20 click on ADD RULES and select ADD my OWN RULES and RULE GROUPS



STEP23: NOW SELECT IP SET

[AWS WAF](#) > [Web ACLs](#) > Create web ACL

Step 1

[Describe web ACL and associate it to AWS resources](#)

Step 2

Add my own rules and rule groups

Step 3

Set rule priority

Step 4

Configure metrics

Step 5

Review and create web ACL

Add my own rules and rule groups [Info](#)

Rule type

Rule type

☐ IP set

Use IP sets to identify a specific list of IP addresses.

☒ Rule builder

Use a custom rule to inspect for patterns including query strings, headers, countries, and rate limit violations.

☐ Rule group

Use a rule group to combine rules into a single logical set.

Rule builder

Rule visual editor

Rule JSON editor

You can use the JSON editor for complex statement nesting, for example to nest two OR statements inside an AND statement. The visual editor handles one level of nesting. For web ACLs and rule groups with complex nesting, the visual editor is disabled.

Rule

Validate

Rule

Name

block-my-laptop-access

The name must have 1-128 characters. Valid characters: A-Z, a-z, 0-9, - (hyphen), and _ (underscore).

IP set

IP set

Choose IP Set

my-laptop-ip

IP address identifies the originating address. When a request comes through a CDN or other proxy network, the source IP address identifies the proxy and the original IP address is sent in a header. Use caution with the option, IP address in header, because headers can be handled inconsistently by proxies and they can be modified to bypass inspection.

☒ Source IP address

☐ IP address in header

Action
Choose an action to take when a request originates from one of the IP addresses in this IP set.

☐ Allow

☒ Block

☐ Count

☐ CAPTCHA

☐ Challenge

STEP24: ONCE DONE CLICK ON ADD RULE

Cancel **Add rule**

STEP25: Once you click on ADD rule it will take you to another page

Step 1
[Describe web ACL and associate it to AWS resources](#)

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Step 5
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Add rules and rule groups [Info](#)

A rule defines attack patterns to look for in web requests and the action to take when a request matches the patterns. Rule groups are reusable collections of rules. You can use managed rule groups offered by AWS and AWS Marketplace sellers. You can also write your own rules and use your own rule groups.

Rules (1) [Edit](#) [Delete](#) [Add rules ▼](#)

If a request matches a rule, take the corresponding action. The rules are prioritized in order they appear.

☐	Name	Capacity	Action
☐	block-my-laptop-access	1	Block

Web ACL capacity units (WCUs) used by your web ACL

The WCUs used by the web ACL will be less than or equal to the sum of the capacities for all of the rules in the web ACL.

The total WCUs for a web ACL can't exceed 5000. Using over 1500 WCUs affects your costs. [AWS WAF Pricing](#)

1/5000 WCUs

SELECT THE CHECKBOX

Step 1

[Describe web ACL and associate it to AWS resources](#)

Step 2

Add rules and rule groups

Step 3

Set rule priority

Step 4

Configure metrics

Step 5

Review and create web ACL

Add rules and rule groups [Info](#)

A rule defines attack patterns to look for in web requests and the action to take when a request matches the patterns. Rule groups are reusable collections of rules. You can use managed rule groups offered by AWS and AWS Marketplace sellers. You can also write your own rules and use your own rule groups.

Rules (1/1)

EditDeleteAdd rules ▼

If a request matches a rule, take the corresponding action. The rules are prioritized in order they appear.

☒	Name	Capacity	Action
☒	block-my-laptop-access	1	Block

Web ACL capacity units (WCUs) used by your web ACL

The WCUs used by the web ACL will be less than or equal to the sum of the capacities for all of the rules in the web ACL.

The total WCUs for a web ACL can't exceed 5000. Using over 1500 WCUs affects your costs. [AWS WAF Pricing](#)

1/5000 WCUs

Default web ACL action for requests that don't match any rules

Default action

☒ Allow

☐ Block

[▶ Custom request - optional](#)

STEP26:ONCE DONE CLICK ON NEXT

AWS WAF > Web ACLs > Create web ACL

Step 1

[Describe web ACL and associate it to AWS resources](#)

Step 2

[Add rules and rule groups](#)

Step 3

Set rule priority

Step 4

Configure metrics

Step 5

Review and create web ACL

Set rule priority [Info](#)

Rules (1)

▲ Move up▼ Move down

If a request matches a rule, take the corresponding action. The rules are prioritized in order they appear.

	Name	Capacity	Action
☐	block-my-laptop-access	1	Block

Cancel

Previous

Next

SELECT THE RADIO BUTTON and click next

The screenshot shows the 'Set rule priority' step in the AWS WAF console. The left sidebar contains a navigation menu with five steps: Step 1 (Describe web ACL and associate it to AWS resources), Step 2 (Add rules and rule groups), Step 3 (Set rule priority), Step 4 (Configure metrics), and Step 5 (Review and create web ACL). The main content area is titled 'Set rule priority' with an 'Info' icon. It features a table with one rule, 'block-my-laptop-access', which has a capacity of 1 and an action of 'Block'. Above the table are 'Move up' and 'Move down' buttons. Below the table are 'Cancel', 'Previous', and 'Next' buttons. A green circle highlights the radio button next to the rule name in the table.

[AWS WAF](#) > [Web ACLs](#) > Create web ACL

Step 1
[Describe web ACL and associate it to AWS resources](#)

Step 2
[Add rules and rule groups](#)

Step 3
Set rule priority

Step 4
Configure metrics

Step 5
Review and create web ACL

Set rule priority [Info](#)

Rules (1/1) [Move up](#) [Move down](#)

If a request matches a rule, take the corresponding action. The rules are prioritized in order they appear.

Name	Capacity	Action
☒ block-my-laptop-access	1	Block

[Cancel](#) [Previous](#) [Next](#)

STEP27:once you click on next it will take you to another page

The screenshot shows the 'Configure metrics' step in the AWS WAF console. The left sidebar is the same as the previous step. The main content area is titled 'Configure metrics' with an 'Info' icon. It has two sections: 'Amazon CloudWatch metrics' and 'Request sampling options'. In the first section, the rule 'block-my-laptop-access' is selected, and its name is entered in the 'CloudWatch metric name' field. In the second section, the 'Enable sampled requests' radio button is selected. At the bottom are 'Cancel', 'Previous', and 'Next' buttons.

[AWS WAF](#) > [Web ACLs](#) > Create web ACL

Step 1
[Describe web ACL and associate it to AWS resources](#)

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[Add rules and rule groups](#)

Step 3
[Set rule priority](#)

Step 4
Configure metrics

Step 5
Review and create web ACL

Configure metrics [Info](#)

Amazon CloudWatch metrics
CloudWatch metrics allow you to monitor web requests, web ACLs, and rules.

Rules	CloudWatch metric name
☒ block-my-laptop-access	block-my-laptop-access

Request sampling options
If you disable request sampling, you can't view requests that match your web ACL rules.

Options

☒ Enable sampled requests
☐ Disable sampled requests
☐ Enable sampled requests with exclusions

[Cancel](#) [Previous](#) [Next](#)

CLICK NEXT verify all the details and click on CREATE WEB ACL

The screenshot shows the 'Step 4: Configure metrics' configuration page in the AWS WAF console. At the top, there is a 'Name' field which is currently empty, with a message 'No items' and 'No items to display' below it. The main section is titled 'Step 4: Configure metrics' and includes an 'Edit step 4' button. Under the heading 'Amazon CloudWatch metrics (1)', there is a table with two columns: 'Rules' and 'CloudWatch metric name'. The table contains one row with 'block-my-laptop-access' in both columns. Below this, the 'Sampled requests' section shows two toggle switches, both of which are currently 'Enabled'. At the bottom of the page, there are three buttons: 'Cancel', 'Previous', and 'Create Web ACL' (highlighted in orange).

Rules	CloudWatch metric name
block-my-laptop-access	block-my-laptop-access

Sampled requests	Sampled requests for web ACL default actions
Enabled	Enabled

STEP28: WEB ACL HAS BEEN CREATED

The screenshot shows the AWS WAF console interface. A green success banner at the top states: 'Success You successfully created the web ACL aws-waf-demo-ec2.' The left sidebar shows the 'WAF & Shield' navigation menu with options like 'Getting started', 'Web ACLs', 'Bot control dashboard', 'Application integration', 'IP sets', 'Regex pattern sets', 'Rule groups', and 'AWS Marketplace managed rules'. The main content area is titled 'Web ACLs' and shows a list of 'Web ACLs (1)' for the 'Europe (Frankfurt)' region. A table lists the created Web ACL:

Name	Description	ID
aws-waf-demo-ec2	-	9d679674-05a4-4595-ae97-04e4a59b8d61

Buttons for 'Copy ARN', 'Delete', and 'Create web ACL' are visible at the top of the Web ACLs list.

STEP29: ONCE DONE verify all the details, click on ASSOCIATED WEB

resources can check if load balances is associated or not and check the rules

WAF & Shield

AWS WAF

- Getting started
- Web ACLs
- Bot control dashboard
- Application integration **New**
- IP sets
- Regex pattern sets
- Rule groups
- AWS Marketplace managed rules

aws-waf-demo-ec2

Download web ACL as JSON

Overview Rules Bot Control **Associated AWS resources** Custom response bodies Logging and metrics

Associated AWS resources (1)

Find associated AWS resources

☐	Name	Resource type	Region
☐	lb-aws-waf-demo	Application Load Balancer	Europe (Frankfurt)

